Graph Structure as Predictive Signal in Bitcoin Money-Laundering Detection

Abstract

As banks and investment firms are setting up settlement infrastructure on blockchains, it has become necessary for cryptocurrency networks to be capable of reliably identifying fraudulent and laundered activity on a large scale **[2][3]**. This project is a supervised binary classification study based on the Elliptic Bitcoin Dataset **[1]**, which is a real-world transaction graph consisting of 203,769 Bitcoin transactions and 234,355 payment relationships spanning 49 time steps, the transactions being labelled as licit, illicit, or unknown. The primary research question is whether the transaction graph contains predictive information that extends beyond that already available in the node-level features and whether this information stays stable or deteriorates over time as illicit methods evolve. In order to answer this question, we design a modular pipeline that compares a simple majority-class benchmark, three progressively more sophisticated feature-based baseline models, and two graph-aware models: a Graph Convolutional Network (GCN) and a Graph Attention Network **[4][5]**. We select F1 and PR-AUC for the illicit class as our main performance measures since there is approximately a 2% class imbalance **[6]**, assess the variability across at least three different random seeds, and carry out three specific studies: a feature-vs-graph ablation, a temporal-robustness study that examines how performance decreases across the later time steps, and a subgroup analysis according to transaction degree. Our contribution is a fully documented, seed-averaged pipeline that shows the additional value offered by the graph structure and illustrates how this value changes over time, an aspect which has not been consistently addressed in previous work **[6][7].**

Research Background

Since 2018, research on anti-money laundering measures and Bitcoin has undergone rapid development. Bartoletti and others have used data mining to identify Bitcoin Ponzi schemes **[2]**, whilst Harley and his colleagues have used supervised learning to de-anonymize addresses on Bitcoin. In 2019, Elliptic released the Elliptic Bitcoin Dataset **[1]**, which became a standard benchmark for graph-based ML classification. Later studies have dealt with comparative supervised methods **[6]** and techniques designed to address the issue of label scarcity **[7]**. At the same time, the graph learning community has developed more sophisticated architectures, for example inductive frameworks such as GraphSAGE **[4]** and attention-based aggregation methods such as Graph Attention Networks **[5]**.

There are still some gaps in the published literature. A number of the proposed solutions give only the overall accuracy or the F1 score without taking into account the approximately 2% class imbalance, a factor which makes the accuracy figure misleading **[6]**. Baseline comparisons are often omitted or restricted to just one classical model, so it is not clear how much of the predictive value is actually due to the graph structure as opposed to more sophisticated feature-only approaches **[7]**. Time-based splits are being applied in an appropriate manner. Only a few studies also examine

performance declines over time steps as malicious actors adjust their methodologies. Reproducibility is likewise poor since the choices regarding preprocessing, the random seeds, and the hyperparameters are seldom completely documented.

This is a problem that deserves investigation, because reliable fraud detection is a necessary precondition for cryptocurrency to function as a reliable part of the mainstream financial system. A detailed and reproducible comparison of strong feature-only baseline methods with graph-aware models directly addresses the question of whether the additional complexity involved in graph learning is justified, which in turn has practical implications for actual AML deployment. Moreover, the project broadens the use of basic ML techniques (such as regularized linear models, tree ensembles, cross-validation, the selection of evaluation metrics, and dealing with imbalanced classification) and neural architectures (GCN, GAT) besides the curriculum to a real-world financial forensics issue.

Methodology

Data: Elliptic Bitcoin Dataset from Kaggle **[1]** (https://www.kaggle.com/datasets/ellipticco/elliptic-data-set). 203,769 transactions and 234,355 payment relationships across 49 time steps, with 166 anonymized numerical features per transaction (94 local + 72 aggregated neighborhood features). Labels: class 1 = illicit (~2%), class 2 = licit (~21%), unknown (~77%). Licensed under CC BY-NC-ND 4.0 for non-commercial academic use with attribution.

Methods:

- **Naive benchmark:** majority-class predictor (Always will guess “licit)
- **Feature-only baselines (ignore graph):** Logistic regression with L2 regularization and class-weighted less, Random Forest, and gradient-bosted trees
- **Graph Neural Network (GNN) models:** two-layer Graph Convolutional Network (CGN) and two-layer Graph Attention Network (GAT) **[4][5]**, both of which use message passing to different features from a node’s neighbors
- **Preprocessing**: Remove unlabeled nodes from supervised training but retain in the for GNN message passing, standardize features using training-partition statistics only, construct PyTorch-Geometric edge index from the edge list

Experiments and Evaluation: All the models will be trained under a strict time-based split where steps 1-34 will be used for training and 35-49 for testing, as was done by previous work ML on this dataset **[6][7]**, with hyperparameters tuned via stratified k-fold cross-validation within the training set and the test set reserved for evaluation only. At least three different random seeds will be used for each model in order to allow calculating the average and standard deviation of the metric. In addition, we plan to perform three ablations: a feature vs. GNN ablation for assessing the effect of graphs, temporal robustness ablation measuring degradation of performance with time, and a degree-based subgroup analysis in order to test whether GNNs help in classifying highly connected transactions. Error analysis will be done focusing on false negatives; missed illicit transactions, which come with the greatest cost in terms of AML. The key evaluation metrics include F1 score and PR-AUC for the illicit class along with precision, recall, and confusion matrix. Accuracy will not be used as the main metric due to the extreme class imbalance

because the “predict licit for everyone” model would achieve about 98% accuracy while identifying no illicit transactions **[6]**

Significance

This study is necessary because detecting the possibility of fraud, which is verifiable, is a fundamental requirement if the use of cryptocurrency is to be reliable in the financial system. Given the adoption of tokenization and settlement based on blockchain technology by large banks and investment firms, the ability to distinguish legal and illegal activities is important for the purposes of compliance, market confidence, and the feasibility of a tokenized economy on a global level **[3]**. The identification of any suspicious activity by using risk assessment for each transaction will help the operators of the exchange and the AML compliance team to take action against such payments in advance and prevent the laundering of money instead of addressing it after the occurrence.

From an academic perspective, our goal will be to design an approach to determine the proportion of the predictive power which stems from the properties of the transaction network versus what was already known from the node features. The model will give us more insight as to whether or not the additional complexity of using Graph Neural Networks is justified against powerful tabular baselines, as well as a well-documented and temporal process to ensure there is no data leakage, a recurring problem in the current state of the art in the field for this dataset **[6][7]**. The findings may prove valuable to AML compliance departments, cryptocurrency exchanges, blockchain analysis firms, and further studies of graph-based financial forensics.

SMART Timeline

Milestone	**Deliverable**	**Team Meetings**	**Final Deadline**
M1: Proposal	Team contract, idea generation, report	Sept 13, 2026	Sept 18, 2026
M2: Dataset & Problem Approval	Source/license, data card, EDA, split strategy, primary metric, feasibility plan	Sept 20 & 27, 2026	Sept 30, 2026
M3: Baseline Reproduction	Working data pipeline, naive benchmark, course-level baseline, reproducible environment, discrepancy analysis	Oct 4 & 11, 2026	Oct 16, 2026
M4: Progress Presentation	Current methods, theoretical justification, preliminary results, blockers, midpoint peer check-in	Oct 18 & 25, 2026	Oct 28, 2026
M5: Final Experiments & Analysis	Fair model comparison, tuning, uncertainty, error analysis, ablation/robustness checks	Nov 1 & 8, 2026	Nov 17, 2026

M6: Final Report, Repository & Demo	8-10 min presentation , 6-8 page report, reproducible repo, contribution/AI-use statements	Nov 15 & 22, 2026	Dec 1, 2026
M7: Individual Reflection & Oral Defence	1-2 page reflection, contribution evidence, oral/code walkthrough	Nov 29 & Dec 6, 2026	Dec 4, 2026

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